

## B.4 Lab 4: Linear Systems and Image Reconstruction

### Problem 1: Mixing an Intravenous Nutrient Infusion

Patients who cannot eat or absorb food through the digestive tract can receive their complete daily nutrition intravenously, a therapy known as total parenteral nutrition.<sup>8</sup> The infusion is prepared by combining commercial stock bags, where each bag type contains a fixed recipe of the three macronutrients: dextrose (sugar), amino acids (protein), and lipids (fat). Our hospital pharmacy stocks three bag types with the following compositions:

|                   | Bag 1 | Bag 2 | Bag 3 |
|-------------------|-------|-------|-------|
| Dextrose (g/L)    | 100   | 50    | 0     |
| Amino acids (g/L) | 20    | 50    | 10    |
| Lipids (g/L)      | 0     | 10    | 100   |

A physician prescribes the following daily totals for a patient: 125 g of dextrose, 50 g of amino acids, and 55 g of lipids. The pharmacist must determine the volume of each bag type to combine so that the mixture delivers the prescription exactly. Let  $x_1$ ,  $x_2$ , and  $x_3$  denote the volumes (in liters) drawn from Bags 1, 2, and 3, respectively. Any choice of volumes delivers some total amount of each macronutrient. For example, the choice  $x_1 = 0.5$ ,  $x_2 = 1$ , and  $x_3 = 0.1$  delivers

$$100(0.5) + 50(1) + 0(0.1) = 100 \text{ g of dextrose,}$$

and computing the amino acid and lipid totals the same way yields 61 g of amino acids and 20 g of lipids. This mixture is not the prescription.

**Problem 1.1:** Determine the volumes that deliver the prescription exactly. To do so, formulate a linear system  $A\mathbf{x} = \mathbf{b}$  whose solution is the correct set of volumes. Then, write a script `tpnPlanner.m` that constructs your  $A$  and  $\mathbf{b}$ , solves the system using MATLAB's backslash operator `\`, and displays the recovered volumes. In your header, describe how you formulated the linear system and state the recovered volumes. To confirm that the computed volumes actually deliver the prescription, your script should also compute and display  $A*\mathbf{x}$ , the macronutrient totals delivered by the recovered volumes.

**Problem 1.2:** The manufacturer of Bag 3 reports a backorder, and the supplier offers a substitute bag whose composition is an equal-parts blend of Bag 1 and Bag 2: 75 g/L dextrose, 35 g/L amino acids, and 5 g/L lipids. Copy your script to a new file `tpnPlannerSub.m`, replace the third column of  $A$  with the substitute composition, and re-run the solve. What does MATLAB report? In your header, explain in a sentence or two why no combination of these three bags can meet the prescription.

### Problem 2: Image Reconstruction & Linear Systems

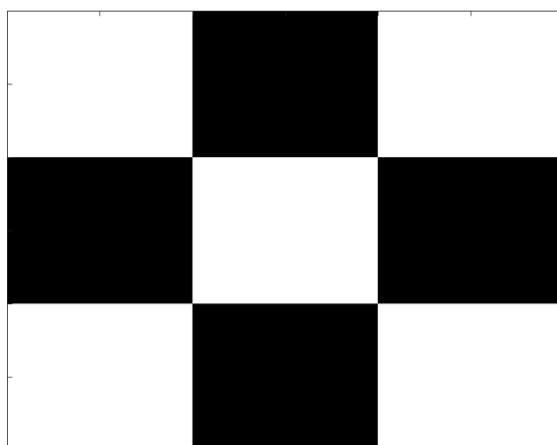
Computed Tomography imaging relies on the beautiful mathematics of the Radon Transform.<sup>9</sup> MATLAB has its own built-in functions to compute the Radon Transform and its inverse,<sup>10</sup> but today we are going to write our own version of a CT reconstruction software in order to image an unknown object.

<sup>8</sup>[https://en.wikipedia.org/wiki/Parenteral\\_nutrition](https://en.wikipedia.org/wiki/Parenteral_nutrition)

<sup>9</sup>[https://en.wikipedia.org/wiki/Radon\\_transform](https://en.wikipedia.org/wiki/Radon_transform)

<sup>10</sup><https://www.mathworks.com/help/images/radon-transform.html>

The general concept of CT is to recover a volumetric image from a series of x-ray projections. In this exercise, we will use a series of one dimensional “x-ray” measurements to recover a 2D image. Let’s take a look at some examples. Here’s a 3x3 image:



Images are nothing more than a matrix of numbers. You can recreate the above image using the following commands:

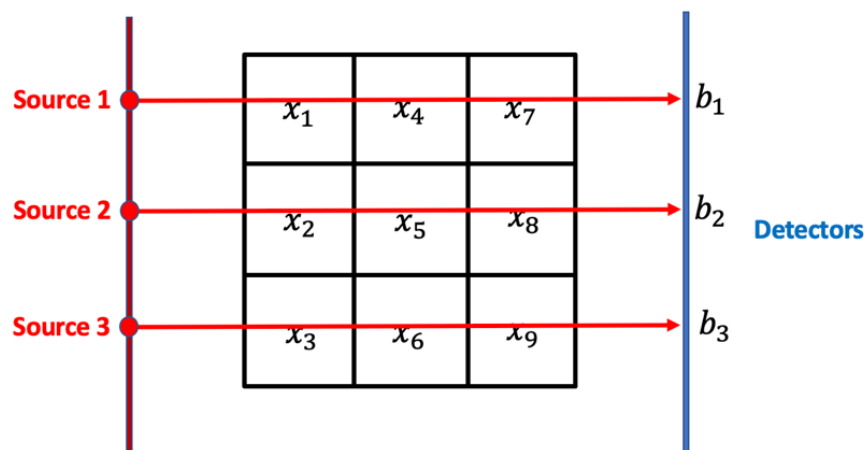
```
>> P = [1 0 1; 0 1 0; 1 0 1]; imagesc(P); colormap('gray')
```

In grayscale, a value of 1 corresponds to white and a value of 0 corresponds to black (shades of gray are values between 0 and 1). Take a moment to verify for yourself that the grayscale image matches up with the values in the matrix  $P$ .

Let’s imagine that we have a new object in the scanner. Our job is to create an image of it. Every pixel value in the image is unknown, but the values are assumed to be constant within their corresponding pixel area. We will use data to define a linear system in order to recover the unknown pixel values. First, let’s label our unknowns:

|       |       |       |
|-------|-------|-------|
| $x_1$ | $x_4$ | $x_7$ |
| $x_2$ | $x_5$ | $x_8$ |
| $x_3$ | $x_6$ | $x_9$ |

For a 3x3 image, there are a total of 9 unknown pixel values, which we label  $x_1, x_2, \dots, x_9$ . Reconstruction data acquired from our scanner comes in the form of 1-D “x-ray” measurements like these:



X-rays (red arrows) act like line integrals. In other words, the value  $b_i$  that the detector receives from source  $i$  can be approximated by the sum of the pixel values that the  $i$ th ray passes through:

$$x_1 + x_4 + x_7 = b_1$$

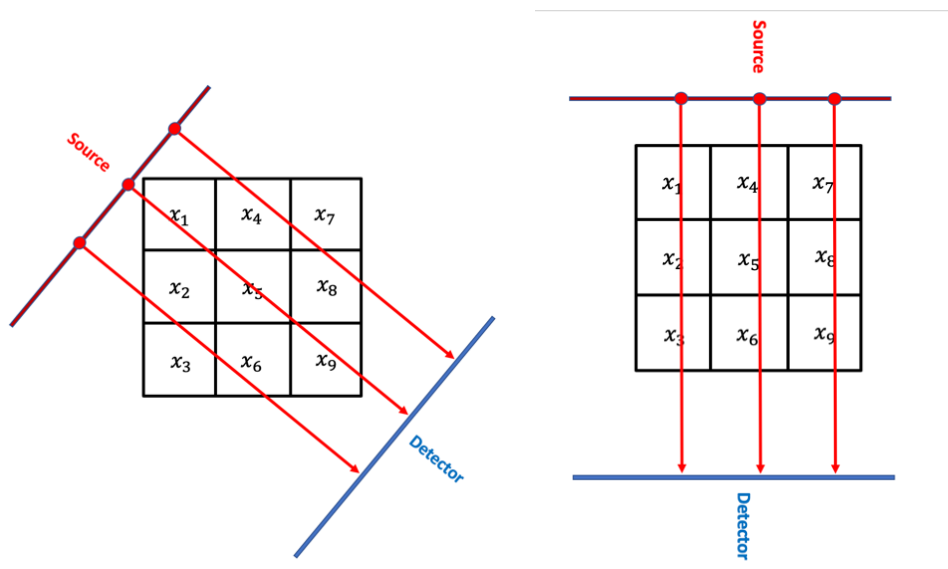
$$x_2 + x_5 + x_8 = b_2$$

$$x_3 + x_6 + x_9 = b_3$$

The data provides three linear equations for the 9 unknown pixel values. We can encode this information in matrix format as:

```
>> A = zeros(9,9);
>> A(1,1) = 1; A(1,4) = 1; A(1,7) = 1;
>> A(2,2) = 1; A(2,5) = 1; A(2,8) = 1;
>> A(3,3) = 1; A(3,6) = 1; A(3,9) = 1;
```

We will need at least six more data points if we want to recover all nine unknowns. So, as is done on a real CT scanner, we rotate the source and detector panels:



Now we would have enough linear equations to determine all nine unknown pixel values. However, I should point out that some care must be taken when approximating the line integrals, but we will

leave that for another day. The main takeaway is that we can represent our image reconstruction problem as a linear system of equations.

CT reconstruction is of course slightly more complicated than this simple algebraic formulation. Try taking a look at the Wikipedia page,<sup>11</sup> or better yet, take Dr. Yankeelov's BME 357 Biomedical Imaging Modalities course to learn more.

**Problem 2.1:** Download `CTdata5x5.p`. This function creates the x-ray data needed to reconstruct an unknown 5 x 5 image. It is called as follows:

```
>> [I,J,V,b] = CTdata5x5;
```

As in the 3x3 example, you will need to setup the matrix  $A$  to encode our reconstruction linear system. The vectors  $I$ ,  $J$ , and  $V$  contain the non-zero entries in the matrix  $A$ . We can use these vectors to populate  $A$ . For instance, the first value in  $V$  is placed into  $A$ , according to the first values in  $I$  and  $J$ :

```
>> A( I(1), J(1) ) = V(1);
```

Once the linear system is built, recover  $\mathbf{x}$  by solving

$$A\mathbf{x} = \mathbf{b}$$

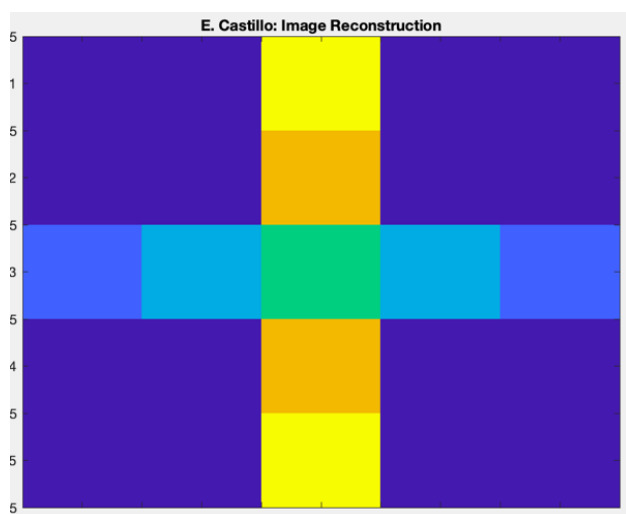
using MATLAB's `\` backslash operator. Note that the right-hand side vector  $\mathbf{b}$  is also returned by `CTdata5x5`. Everything that needs to be done is outlined in the incomplete function `imageRecon5x5.m`. Download it and complete it for this problem.

**Problem 2.2:** Having successfully solved for the vector  $\mathbf{x}$ , you will need to format the solution into an image,  $P$ . The vector  $\mathbf{x}$  is organized in the same manner as the previous 3x3 example:

|       |          |          |          |          |
|-------|----------|----------|----------|----------|
| $x_1$ | $x_6$    | $x_{11}$ | $x_{16}$ | $x_{21}$ |
| $x_2$ | $x_7$    | $x_{12}$ | $x_{17}$ | $x_{22}$ |
| $x_3$ | $x_8$    | $x_{13}$ | $x_{18}$ | $x_{23}$ |
| $x_4$ | $x_9$    | $x_{14}$ | $x_{19}$ | $x_{24}$ |
| $x_5$ | $x_{10}$ | $x_{15}$ | $x_{20}$ | $x_{25}$ |

Write code to place the recovered  $\mathbf{x}$  values into their correct locations in  $P$ , then create a figure illustrating your reconstructed image using MATLAB's `imagesc` function. Be sure to include a title on the figure with your name. It should look something like this:

<sup>11</sup>[https://en.wikipedia.org/wiki/Tomographic\\_reconstruction](https://en.wikipedia.org/wiki/Tomographic_reconstruction)



## General Submission Guidelines

|                                                          |                                              |
|----------------------------------------------------------|----------------------------------------------|
| <code>[FirstName][LastName]_[filename].m</code>          | For programs                                 |
| <code>[FirstName][LastName]_[filename]_plot_1.pdf</code> | For corresponding plots (export them to pdf) |
| <code>[FirstName][LastName]_[filename]_plot_2.pdf</code> | For problem with more than one plot          |

## Submission Checklist

- Problem 1.1: `tpnPlanner.m`
  - Complete header (system formulation described, recovered volumes stated) and commented code
  - Correct formulation and construction of  $A$  and  $\mathbf{b}$
  - Correct use of MATLAB's backslash operator and delivered-totals check
- Problem 1.2: `tpnPlannerSub.m`
  - Correct substitute matrix
  - Header explanation of why the prescription cannot be met
- Problem 2.1: `imageRecon5x5.m`
  - Fully commented code and header
  - Correct construction of the matrix  $A$
  - Correct use of MATLAB's backslash operator to solve linear system
- Problem 2.2
  - Correct indexing into  $P$
  - Correct image figure